

Axiomatic Necessity and External Structural Support in Machine Learning Systems

Abstract

1. Introduction
2. Theorem of Axiomatic Necessity (Brief Recap)
3. Mapping TNA to Machine Learning
4. Why Data Alone Cannot Ground Learning
5. Inductive Bias as Necessary Axiom
6. Generalization as External Support, Not Emergence
7. Foundation Models and the Illusion of Closure
8. Consequences
 - 8.1 No Fully Self-Sustaining Learning System
 - 8.2 Externality Is Not a Flaw
 - 8.3 Collapse of the “Purely Data-Driven” Myth
9. Relation to Other Results
10. Conclusion

Axiomatic Necessity and Inductive Bias:

Abstract

1. Introduction
2. Preliminaries
 - 2.1 Learning Systems
 - 2.2 Generalization
3. Theorem of Axiomatic Necessity (TNA)
4. Observational Closure in Learning Systems
5. Ontological Incompleteness of Learning Systems
 - Theorem 1 (Ontological Incompleteness of Learning Systems)
 - Proof (Sketch)
6. Inductive Bias as Axiomatic Necessity
7. Relation to No Free Lunch Theorems
8. Foundation Models and Hidden Axioms
9. Implications
10. Conclusion

Acknowledgments

Appendix A: Implicit Bias in Stochastic Gradient Descent as Axiomatic Necessity

- A.1 Empirical Risk Minimization Is Underdetermined
- A.2 SGD Selects a Particular Solution
- A.3 Implicit Bias Is Not Data-Derivable
- A.4 Implicit Bias as Necessary Axiom
- A.5 Structural Interpretation
- A.6 Consequences
- A.7 Summary

Appendix B: Double Descent as a Symptom of Axiomatic Tension

- B.1 Classical Bias–Variance Tradeoff and Its Breakdown
- B.2 Interpolation and Underdetermination

B.3 Structural Selection Beyond Interpolation

B.4 Double Descent as Axiomatic Tension

B.5 Why Double Descent Is Not Accidental

B.6 Implications

B.7 Summary

Discussion: Implications for Interpretability and Alignment

Interpretability as Axiom Identification

Alignment Depends on Structural Transparency

Why Scaling Does Not Resolve These Issues

Structural Honesty as a Design Principle

Summary

Anticipated Reviewer Objections and Responses

Objection 1: “This is just inductive bias under a new name.”

Objection 2: “These ideas are already well-known in statistical learning theory.”

Objection 3: “The paper is philosophical rather than technical.”

Objection 4: “There is no new algorithm or empirical result.”

Objection 5: “This does not change practical model development.”

Objection 6: “Double descent has already been explained in other ways.”

Objection 7: “The framework seems unfalsifiable.”

Objection 8: “This work risks overgeneralization.”

Closing Remark for Reviewers

Why This Work Belongs at NeurIPS

Relevance to Core NeurIPS Themes

Learning Beyond Interpolation

Optimization as Structure, Not Just Computation

Interpretability and Alignment

Contribution Type: Foundational Clarification

Timeliness

Summary

Rebuttal

“This is just inductive bias / implicit regularization.”

“This idea already exists in learning theory / philosophy of science.”

“This is philosophical rather than technical.”

“There is no new algorithm or empirical result.”

“The claims seem too general.”

Clarification Added in Revision

Summary

Original and Substantive Contributions to Machine Learning from the Axiomatic Necessity Theory (TNA)

1. Reframing Inductive Bias as Structural Axiomatic Necessity

2. Proof That No Learning System Can Be Observationally Closed

3. Interpreting Implicit Bias (e.g., in SGD) as Manifestation of Necessary Axioms

4. Double Descent as a Symptom of Axiomatic Tension

5. Structural Critique of Foundation Models
6. Implications for Interpretability and Alignment

Relation to Existing Work

Conclusion

Axiom Ablation Test (AAT): Validating Axiomatic Necessity in Deep Learning

1. Motivation
2. Core Hypothesis (H_1)
3. Experimental Design
 - 3.1 Base Architecture
 - 3.2 Axiomatic Conditions (Independent Variable)
 - 3.3 Controlled Variables
4. Primary Outcome Measures
5. Secondary Analyses
 - 5.1 Axiom Localization via Probing
 - 5.2 Explicit Bias Injection
6. Expected Results (Visual Summary)
7. Statistical Analysis
8. Interpretation Framework
9. Limitations & Controls
10. Broader Impact Statement (para NeurIPS)

Axiomatic Necessity and External Structural Support in Machine Learning Systems

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Abstract

Modern machine learning systems are often presented as self-sufficient: trained end-to-end from data, optimized by internal loss functions, and evaluated solely by empirical performance. This paper argues that such systems are not structurally self-sustaining. Using the **Theorem of Axiomatic Necessity (TNA)**, we show that every coherent learning system necessarily depends on at least one external structural element that cannot be derived from the system itself. In machine learning, these elements appear as inductive biases, architectural constraints, objective functions, and data selection principles. We formalize this dependence and show that attempts to eliminate external structure lead to incoherence, instability, or trivialization. The result reframes generalization not as an emergent statistical miracle, but as the consequence of unavoidable external axioms.

1. Introduction

Machine learning (ML) is frequently described as a paradigm in which structure is *learned from data*. In its strongest form, this narrative suggests that sufficiently large models, trained on sufficient data, can dispense with explicit assumptions about the world.

Despite its rhetorical appeal, this view is structurally false.

This paper demonstrates that **no coherent machine learning system can be axiom-free**, even in principle. Any system that generalizes, converges, or remains stable necessarily relies on external structural constraints that cannot be derived from training data alone.

We show that this is not an empirical limitation, but a **metamathematical necessity**, formalized by the Theorem of Axiomatic Necessity (TNA).

2. Theorem of Axiomatic Necessity (Brief Recap)

Let:

- A be a (possibly infinite) set of axioms or structural assumptions,
- E a set of observable or functional requirements (e.g. correct predictions, convergence, stability).

If

$$A \models E,$$

then there exists a finite minimal subset

$$F_{\min} \subseteq A$$

such that:

1. $F_{\min} \models E$,
2. For every $a \in F_{\min}$,

$$(A \setminus \{a\}) \not\models E.$$

Each element of $F_{\min}$ is **metamathematically necessary**: it cannot be removed without collapse of coherence.

Crucially, such elements **cannot be justified from within the system they sustain** without circularity.

3. Mapping TNA to Machine Learning

We now identify the correspondence between the abstract objects of TNA and concrete ML systems.

TNA CONCEPT	MACHINE LEARNING INTERPRETATION
Axioms A	Architectural choices, loss functions, optimization rules, data priors
Observations E	Generalization, stable training, meaningful representations
$F_{\min}$	Minimal inductive bias required for learning
Removal of a	Training collapse, overfitting, trivial solutions

Machine learning systems are **formal systems** in the precise sense required by TNA: they manipulate symbols, optimize objectives, and produce outputs constrained by formal rules.

4. Why Data Alone Cannot Ground Learning

A common claim is that *data replaces axioms*. This is false for structural reasons.

Training data itself presupposes:

- a representation space,
- a similarity metric,
- a loss function,
- a sampling process,
- an optimization dynamic.

None of these are derivable from the dataset itself. They are **external supports**.

If one attempts to remove all such supports, the system becomes underdetermined:

- any function interpolates the data,
- generalization is undefined,
- learning degenerates into memorization.

This is an instance of **reduction to incoherence**, exactly as described by the TNA.

5. Inductive Bias as Necessary Axiom

Inductive bias is often treated as a design choice. The TNA shows it is **unavoidable**.

Formally:

If a learning system generalizes, then it possesses at least one inductive bias that cannot be learned from data.

This bias may take the form of:

- convolutional locality,

- attention mechanisms,
- smoothness assumptions,
- sparsity constraints,
- architectural symmetries.

Removing all bias does not produce a neutral learner; it produces **no learner at all**.

6. Generalization as External Support, Not Emergence

Generalization is frequently described as an emergent phenomenon. Under the TNA, this interpretation fails.

If generalization were emergent from the system itself, it would be derivable from the training process. But:

- generalization requires exclusion of infinitely many hypotheses,
- exclusion requires criteria,
- criteria require axioms.

Thus, generalization is the *effect* of external structural necessity, not a spontaneous property of data.

7. Foundation Models and the Illusion of Closure

Large-scale models (LLMs, diffusion models, foundation models) intensify the illusion of self-sufficiency.

Yet they rely even more heavily on external structure:

- curated datasets,
- tokenization schemes,
- pretraining objectives,
- architectural priors,
- human-defined evaluation metrics.

Scale does not eliminate axioms; it **hides them**.

8. Consequences

8.1 No Fully Self-Sustaining Learning System

There exists no machine learning system whose coherence is internally justified.

8.2 Externality Is Not a Flaw

The necessity of external structure is not a weakness of ML; it is a structural fact shared by all coherent systems.

8.3 Collapse of the “Purely Data-Driven” Myth

Data-driven does not mean axiom-free. It means **axioms are implicit**.

9. Relation to Other Results

The TNA-based result is structurally aligned with:

- No Free Lunch Theorems (but stronger),
- Gödelian incompleteness (but orthogonal),
- Bias–variance tradeoff (as a symptom, not a cause).

Unlike these, the TNA identifies **why** external structure is unavoidable, not merely that tradeoffs exist.

10. Conclusion

Machine learning systems do not learn *from nothing*. They operate as finite coherent structures sustained by external supports.

The Theorem of Axiomatic Necessity proves that this dependence is not contingent, historical, or technological — it is **structural**.

Ontological completeness is impossible.

But **structural honesty** — the explicit acknowledgment of necessary external support — is not only possible; it is essential for understanding what machine learning truly is.

Axiomatic Necessity and Inductive Bias:

Why Learning Systems Cannot Be Structurally Self-Sustaining**

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January 2026

Abstract

Machine learning systems are often described as data-driven and increasingly self-sufficient. However, despite empirical success, no learning system operates without external structural assumptions. In this work, we formalize this limitation using the **Theorem of Axiomatic Necessity (TNA)**. We show that any learning system capable of generalization necessarily depends on at least one axiom that cannot be derived from data or internal optimization alone. This axiom manifests as inductive bias, architectural constraint, or objective selection. Our result reframes generalization as a consequence of unavoidable external structure rather than emergent statistical behavior, strengthening and extending classical results such as the No Free Lunch theorems.

1. Introduction

A recurring narrative in modern machine learning is that scale and data can replace explicit assumptions. Deep learning architectures are often presented as learning structure directly from data, minimizing the role of prior commitments.

This paper argues that this narrative is **structurally false**.

We prove that **no coherent learning system can be axiomatically closed**. Any system that generalizes beyond its training data must rely on at least one externally imposed structural element that cannot be justified from within the system itself.

Our contribution is not algorithmic but **metatheoretical**: we identify inductive bias as an instance of *axiomatic necessity* and show that its presence is unavoidable in principle.

2. Preliminaries

2.1 Learning Systems

We consider supervised learning systems defined by a tuple

$$\mathcal{L} = (\mathcal{X}, \mathcal{Y}, \mathcal{H}, \ell, \mathcal{A})$$

where:

- $\mathcal{X}$ is the input space,
- $\mathcal{Y}$ the output space,
- $\mathcal{H}$ a hypothesis class,
- ℓ a loss function,
- $\mathcal{A}$ a learning algorithm.

Training data

$$D = \{(x_i, y_i)\}_{i=1}^n$$

is assumed finite.

2.2 Generalization

A system generalizes if performance on unseen data is systematically better than random selection over $\mathcal{H}$.

3. Theorem of Axiomatic Necessity (TNA)

We adopt the following result (Bresciano, 2026):

Theorem (Axiomatic Necessity).

Let S be a coherent formal system that satisfies a nontrivial functional property P . Then there exists at least one axiom a such that:

1. a is necessary for P ,
2. a cannot be derived from the remaining axioms of S ,
3. any attempt to justify a internally leads to circularity.

Such axioms are called **necessary axioms**.

4. Observational Closure in Learning Systems

We define:

Definition 1 (Observational Closure).

A learning system is observationally closed if all hypothesis selection criteria depend exclusively on the observed dataset D .

This corresponds to the idealized notion of a “purely data-driven” learner.

5. Ontological Incompleteness of Learning Systems

We now state the central result.

Theorem 1 (Ontological Incompleteness of Learning Systems)

No observationally closed learning system can satisfy all three of the following simultaneously:

1. Non-arbitrary hypothesis selection,
2. Stability under perturbations of data,
3. Generalization beyond the training set.

Proof (Sketch)

Given finite data D , infinitely many hypotheses in $\mathcal{H}$ are consistent with D . Without additional structure, all such hypotheses are equivalent under empirical risk minimization.

Any rule that selects one hypothesis over others introduces a preference not derivable from D . This preference constitutes an axiom external to the observational domain.

By the TNA, this axiom is necessary and irreducible. Eliminating it collapses the system into arbitrariness or triviality. ■

"Because the result concerns structural necessity rather than algorithmic construction, the proof is given as a reduction-to-incoherence sketch, as is standard for impossibility and underdetermination results."

6. Inductive Bias as Axiomatic Necessity

Inductive bias is often framed as a design choice. Under Theorem 1, it is a **structural necessity**.

Examples include:

- locality in convolutional networks,
- attention constraints in transformers,
- smoothness assumptions in kernel methods,
- sparsity priors in regularization.

These biases cannot be learned from data alone; they must be assumed.

7. Relation to No Free Lunch Theorems

The No Free Lunch (NFL) theorems show that no learner is universally optimal over all possible data-generating distributions.

Our result is stronger:

- NFL identifies a limitation over distributions.
- TNA identifies a **structural necessity** within the learning system itself.

Even restricting the problem class does not eliminate axiomatic dependence; it merely relocates it.

8. Foundation Models and Hidden Axioms

Large-scale models amplify the illusion of self-sufficiency. However, they rely on extensive external structure:

- curated datasets,

- architectural priors,
- tokenization schemes,
- human-defined objectives.

Scale hides axioms but does not remove them.

9. Implications

- **No axiom-free learning:** learning without bias is impossible.
 - **Generalization is not emergent:** it is enforced by structure.
 - **Model comparison requires axiom awareness:** performance differences reflect differing structural commitments.
-

10. Conclusion

We have shown that machine learning systems cannot be structurally self-sustaining. Any system that learns coherently must rely on external axioms that cannot be derived from data alone.

This dependence is not a flaw but a necessity.

Ontological completeness is impossible.

But **ontological clarity**—the explicit acknowledgment of necessary external structure—is both achievable and essential for understanding learning systems.

Acknowledgments

The author thanks the broader ML community for repeatedly rediscovering inductive bias under different names.

Appendix A: Implicit Bias in Stochastic Gradient Descent as Axiomatic Necessity

This appendix provides a concrete technical example illustrating how axiomatic necessity manifests in practical machine learning systems. We focus on the well-studied phenomenon of **implicit bias in stochastic gradient descent (SGD)**.

A.1 Empirical Risk Minimization Is Underdetermined

Consider a supervised learning problem with finite dataset

$$D = \{(x_i, y_i)\}_{i=1}^n$$

and hypothesis class $\mathcal{H}$ parameterized by $\theta \in \mathbb{R}^d$.

For many modern models (e.g. overparameterized neural networks), there exist infinitely many parameter vectors θ such that:

$$\ell(\theta; D) = 0$$

i.e., they interpolate the training data perfectly.

From the perspective of empirical risk minimization alone, all such solutions are equivalent. The learning problem is therefore **structurally underdetermined**.

A.2 SGD Selects a Particular Solution

Despite this underdetermination, stochastic gradient descent consistently converges to **specific subclasses of solutions**, depending on factors such as:

- initialization,
- learning rate schedule,
- batch noise,
- parameterization.

Classic results show that, in linear models, SGD converges to the **minimum Euclidean norm solution**, even when no explicit regularization term is present. Similar norm- or margin-based biases have been identified in deep networks.

This phenomenon is known as **implicit bias**.

A.3 Implicit Bias Is Not Data-Derivable

Crucially, the preference for minimum-norm or maximum-margin solutions:

- is not encoded in the dataset,
- is not specified in the loss function,
- is not enforced by the hypothesis class alone.

It arises from the **optimization dynamics themselves**.

Thus, the optimization algorithm introduces a structural preference that cannot be derived from observed data. This preference functions as an **external axiom** relative to the observational domain.

A.4 Implicit Bias as Necessary Axiom

From the perspective of the Theorem of Axiomatic Necessity:

- The system property P is generalization and stability.
- The dataset D is insufficient to determine a unique solution.
- The implicit bias of SGD selects a specific hypothesis.

Removing this bias—e.g., replacing SGD with a uniformly random selector over interpolating solutions—destroys generalization.

Therefore, the implicit bias of SGD is:

1. **Necessary** for non-arbitrary learning,
2. **Irreducible** to data or loss,
3. **Not internally justifiable** without circularity.

It is an instance of axiomatic necessity.

A.5 Structural Interpretation

Implicit bias is often described as an emergent property of optimization. The present framework suggests a stronger interpretation:

*Optimization dynamics act as **external structural support**, enforcing selection among otherwise equivalent hypotheses.*

SGD does not merely find a solution; it **imposes structure** on the solution space.

This structure is not learned. It is assumed.

A.6 Consequences

This example illustrates the central claim of the paper:

- Even in purely algorithmic settings,
- Even without explicit regularization,
- Even in large-scale models,

learning requires axioms.

Implicit bias is not a nuisance to be eliminated, but a necessary component that sustains coherence.

A.7 Summary

SGD implicit bias provides a concrete, widely accepted example of axiomatic necessity in machine learning. It demonstrates that external structural constraints are not philosophical artifacts, but operational requirements without which learning collapses into arbitrariness.

Appendix B: Double Descent as a Symptom of Axiomatic Tension

This appendix interprets the double descent phenomenon as a structural symptom of unresolved axiomatic tension in learning systems, rather than as an anomaly of model capacity or data scale.

B.1 Classical Bias–Variance Tradeoff and Its Breakdown

In classical statistical learning theory, test error follows a U-shaped curve as a function of model complexity, reflecting a balance between bias and variance.

Recent empirical results in modern machine learning demonstrate a different behavior: as model capacity increases, test error initially increases near the interpolation threshold and then decreases again. This phenomenon is known as **double descent**.

B.2 Interpolation and Underdetermination

At the interpolation threshold, the model becomes capable of fitting the training data exactly. At this point:

- the empirical risk is zero,
- the hypothesis space contains infinitely many solutions consistent with the data,
- empirical evidence alone cannot distinguish between them.

This marks a regime of **maximal underdetermination**.

The spike in test error at this regime reflects not excess capacity per se, but the absence of sufficiently strong structural constraints to resolve this ambiguity.

B.3 Structural Selection Beyond Interpolation

As capacity increases further, optimization dynamics and architectural constraints increasingly dominate hypothesis selection.

Empirically, this corresponds to:

- norm minimization effects,
- margin maximization,
- architectural inductive biases becoming more pronounced.

Generalization improves not because data has become more informative, but because **structural selection has strengthened**.

B.4 Double Descent as Axiomatic Tension

From the perspective of the Theorem of Axiomatic Necessity:

- Below interpolation, the system is partially constrained by data.
- At interpolation, data constraints vanish.
- Beyond interpolation, learning is sustained only by external axioms encoded in architecture and optimization.

The double descent curve thus reflects a transition between:

1. data-constrained selection, and
2. axiom-constrained selection.

The peak of the curve corresponds to a regime where axiomatic structure is insufficiently strong to stabilize hypothesis selection.

B.5 Why Double Descent Is Not Accidental

If learning systems were axiomatically closed, increasing capacity would monotonically degrade generalization once data constraints are exhausted.

The empirical recovery of generalization beyond interpolation indicates the activation of latent structural constraints.

Double descent is therefore not an anomaly—it is evidence that learning systems are **structurally incomplete but axiomatically supported**.

B.6 Implications

This interpretation has several consequences:

- Double descent cannot be eliminated by more data alone.
 - Architectural and optimization choices directly shape generalization regimes.
 - Attempts to “explain away” double descent without reference to structural selection miss its root cause.
-

B.7 Summary

Double descent reveals the fault line between data sufficiency and axiomatic necessity. It is a visible manifestation of the fact that learning systems do not generalize because data compels them to, but because structural constraints make generalization possible.

Discussion: Implications for Interpretability and Alignment

The results presented in this paper have direct consequences for two central concerns in contemporary machine learning: interpretability and alignment.

Interpretability as Axiom Identification

Interpretability is often framed as the task of explaining model decisions in terms of inputs, features, or internal representations. Under the present framework, this framing is incomplete.

If learning systems necessarily depend on external axioms—manifested as inductive biases, architectural constraints, or optimization dynamics—then a significant portion of model behavior is not *learned* but *imposed*.

From this perspective, interpretability should be understood not only as:

- “Why did the model produce this output given the input?”
but also as:
- “Which structural assumptions made this output possible?”

Many interpretability failures arise when explanations are sought entirely within the observational domain (data and activations), while the decisive constraints lie in the latent structural domain (architecture, optimization, and objective design).

Thus, interpretability is fundamentally a problem of **axiom localization**.

Alignment Depends on Structural Transparency

Alignment is commonly treated as a matter of objective specification or reward shaping. However, objectives themselves are axiomatic elements.

Because axiomatic necessity cannot be eliminated, alignment cannot be achieved by optimizing over behavior alone. It requires explicit awareness of:

- which axioms govern hypothesis selection,

- how optimization dynamics amplify or suppress them,
- and which structural biases persist regardless of data.

Misalignment often arises when systems appear aligned empirically but operate under latent axioms that diverge from intended goals.

This explains why alignment failures can occur even when:

- training data is carefully curated,
- objectives are explicitly defined,
- performance metrics are satisfied.

The failure is not behavioral but **structural**.

Why Scaling Does Not Resolve These Issues

Scaling increases expressive capacity and statistical power, but it does not remove axiomatic dependence. On the contrary, scale tends to obscure axioms by distributing their effects across large parameter spaces.

As a result:

- interpretability becomes harder,
- alignment becomes more brittle,
- and post-hoc explanations become less informative.

Recognizing axiomatic necessity reframes these difficulties as inherent rather than incidental.

Structural Honesty as a Design Principle

The present framework suggests a shift in emphasis:

- from eliminating bias to **making bias explicit**,
- from post-hoc explanation to **structural transparency**,
- from behavioral alignment to **axiomatic alignment**.

Alignment, in this view, is not the elimination of external structure, but its deliberate selection.

Summary

Interpretability and alignment cannot be fully addressed within observational closure. Both require acknowledging that learning systems are sustained by necessary external axioms.

What cannot be removed must be understood.

Ontological completeness is impossible.

Structural honesty is not.

Anticipated Reviewer Objections and Responses

Objection 1: “This is just inductive bias under a new name.”

Response:

Inductive bias is typically treated as an empirical or heuristic property of learning algorithms. In contrast, this work formalizes inductive bias as a **structural necessity**, not an optional design choice.

The key distinction is that inductive bias is often described as something that *improves performance*, whereas we show that some form of bias is **logically required** for learning to be well-defined beyond interpolation.

The contribution is therefore not terminological, but foundational: we provide a necessity result explaining *why* inductive bias cannot be eliminated, rather than cataloging *which* biases perform well.

Objection 2: “These ideas are already well-known in statistical learning theory.”

Response:

While many components of the argument—underdetermination, implicit bias, architectural constraints—are individually well established, they are typically treated as contingent phenomena.

This work contributes a unifying structural claim: that learning systems are ontologically incomplete unless supported by external axioms, and that this incompleteness is not accidental but necessary.

To our knowledge, existing treatments describe *effects* (e.g., implicit bias, double descent) without identifying their shared necessity as manifestations of axiomatic dependence.

Objection 3: “The paper is philosophical rather than technical.”

Response:

The paper is metatheoretical, not philosophical. It does not introduce speculative ontology, but formalizes constraints already operative in modern learning systems.

The framework is grounded in:

- concrete optimization dynamics (e.g., SGD implicit bias),
- empirical phenomena (e.g., double descent),
- and standard mathematical notions of underdetermination.

No metaphysical assumptions are introduced beyond what is required to make these phenomena coherent.

Objection 4: “There is no new algorithm or empirical result.”

Response:

The goal of this work is explanatory rather than algorithmic.

NeurIPS regularly includes work that:

- clarifies foundational assumptions,
- reframes known phenomena,
- or exposes hidden constraints affecting algorithm design.

By identifying axiomatic necessity as a structural constraint, this work informs future algorithmic design by making explicit what cannot be learned from data alone.

Objection 5: “This does not change practical model development.”

Response:

On the contrary, recognizing axiomatic necessity has direct implications for:

- architecture selection,
- optimization strategy design,
- interpretability tooling,
- and alignment objectives.

Treating structural assumptions as necessary rather than incidental enables more deliberate and transparent design choices, particularly in high-stakes or safety-critical systems.

Objection 6: “Double descent has already been explained in other ways.”

Response:

Existing explanations of double descent focus on specific mechanisms (e.g., norm minimization, overparameterization effects).

This work does not dispute those mechanisms; it provides a higher-level explanation of why such mechanisms must exist.

Double descent is interpreted not as a capacity anomaly, but as a transition between data-constrained and axiom-constrained learning regimes.

Objection 7: “The framework seems unfalsifiable.”

Response:

The framework makes falsifiable claims at the structural level.

For example, it predicts that:

- learning without external structural constraints cannot generalize beyond interpolation,
- removing or weakening axiomatic support should increase instability and variance,
- strengthening structural constraints should improve generalization even with fixed data.

These predictions are testable via controlled ablations of architecture and optimization.

Objection 8: “This work risks overgeneralization.”

Response:

The claims are deliberately narrow.

We do not assert which axioms are optimal, universal, or correct—only that some axioms are necessary for coherence.

The framework constrains what is possible; it does not prescribe specific implementations.

Closing Remark for Reviewers

This paper does not aim to replace existing theory, but to make explicit a structural dependency that is already implicitly assumed.

What is not acknowledged cannot be controlled.

What is necessary should be made visible.

Why This Work Belongs at NeurIPS

Modern machine learning has entered a regime where performance gains are increasingly driven by scale, architecture, and optimization dynamics rather than by novel data sources or handcrafted features. In this regime, foundational assumptions that were previously treated as implementation details have become central determinants of behavior, generalization, and failure modes.

This paper addresses a structural question that lies beneath many current debates in machine learning:

What must be assumed—independently of data—for learning to be possible at all?

Relevance to Core NeurIPS Themes

Learning Beyond Interpolation

As models enter heavily overparameterized regimes, classical explanations of generalization based purely on data become insufficient. Phenomena such as implicit bias, norm minimization, and double descent demonstrate that learning outcomes depend critically on constraints not specified by training data.

This work provides a unifying explanation: learning systems necessarily rely on external axioms—structural constraints that cannot be learned from data but are required for coherent hypothesis selection.

This reframing clarifies why modern models generalize despite extreme underdetermination.

Optimization as Structure, Not Just Computation

NeurIPS has seen increasing focus on optimization dynamics (e.g., SGD, adaptive methods) as implicit regularizers. This paper places those observations within a necessity framework: optimization algorithms do not merely optimize; they *instantiate axioms* that resolve indeterminacy.

Understanding optimization as a source of necessary structure rather than a neutral solver aligns directly with current research directions.

Interpretability and Alignment

Interpretability and alignment are increasingly recognized as structural problems rather than purely empirical ones. Post-hoc explanations often fail because decisive constraints lie outside the observable data domain.

By formalizing axiomatic dependence, this work provides a principled explanation for why interpretability is difficult and why alignment cannot be achieved through behavioral optimization alone.

This perspective complements ongoing NeurIPS efforts in explainable AI and AI safety.

Contribution Type: Foundational Clarification

NeurIPS has a long tradition of publishing work that reshapes how existing results are understood, even when no new algorithm is introduced.

This paper contributes:

- a necessity result rather than a performance claim,
- a unifying explanation rather than a mechanism-specific analysis,
- and a structural lens applicable across architectures, datasets, and training regimes.

Such contributions are essential for guiding future research directions in a rapidly scaling field.

Timeliness

As machine learning systems become:

- larger,
- less interpretable,
- and more socially consequential,

the cost of unexamined assumptions increases.

Making axiomatic dependence explicit is not a philosophical exercise; it is a prerequisite for responsible and effective system design.

Summary

This work belongs at NeurIPS because it addresses a question that modern machine learning can no longer afford to ignore:

**Learning systems do not stand on data alone.
What supports them should be made explicit.**

Rebuttal

We thank the reviewers for their careful reading and constructive feedback. Below we address the main concerns raised, clarifying both the scope and the contribution of the paper.

“This is just inductive bias / implicit regularization.”

We agree that inductive bias, implicit regularization, and architectural constraints are well-known concepts in machine learning. However, the contribution of this paper is **not** to reintroduce these notions, but to establish a **necessity result** about them.

Most existing work treats inductive bias as a contingent design choice: something that can be tuned, reduced, or traded off. Our central claim is stronger and orthogonal:

■ *Some external structural constraint is necessary for learning to be coherent at all.*

In other words, inductive bias is not merely helpful—it is unavoidable. The paper formalizes this claim by showing that any learning system that purports to be closed under its own hypotheses must implicitly rely on axiomatic elements that cannot be derived from data alone.

This moves the discussion from *which bias is better* to *why some bias must exist*. To our knowledge, this necessity framing is absent from the inductive bias literature, which is primarily comparative and empirical rather than structural.

“This idea already exists in learning theory / philosophy of science.”

Elements related to our argument indeed appear across multiple literatures:

- No Free Lunch theorems,
- implicit bias analyses of SGD,
- minimum norm solutions,
- Occam-style arguments,
- philosophical discussions of foundational assumptions.

However, these results are typically **local**, addressing specific algorithms or regimes, and do not articulate a unified necessity claim.

Our contribution is to:

1. isolate the shared structural assumption across these results,
2. formalize it as axiomatic necessity rather than algorithmic preference,
3. and connect it explicitly to modern phenomena such as overparameterization and double descent.

The value of the paper lies precisely in making explicit what is otherwise scattered, implicit, or treated as incidental.

This is consistent with NeurIPS’ tradition of publishing unifying and clarifying foundational work.

“This is philosophical rather than technical.”

The paper is theoretical, but not speculative. All claims are grounded in standard learning-theoretic settings and refer to concrete phenomena (SGD dynamics, model selection under underdetermination, generalization without explicit regularization).

The key distinction is between:

- **epistemic explanations** (what we observe),
- and **structural explanations** (what must be assumed).

The latter are rarely articulated explicitly in ML, yet they govern system behavior in practice. Making these assumptions explicit is a technical contribution because it constrains what explanations are valid and what questions are meaningful.

We emphasize that the paper does not propose new metaphysical entities or unverifiable constructs. “External axioms” refer concretely to architectural, algorithmic, and optimization constraints already present in deployed systems.

“There is no new algorithm or empirical result.”

This is correct by design. The contribution is **foundational**, not algorithmic.

NeurIPS has a long history of accepting papers that:

- clarify theoretical assumptions,
- reinterpret existing results,
- or expose hidden structure underlying empirical success.

Given the current state of the field—where scaling dominates and interpretability and alignment are pressing concerns—we believe clarifying the necessary conditions of learning is timely and relevant.

Moreover, the framework provides a principled lens through which existing empirical results (e.g., double descent, implicit bias) can be better understood, even if it does not introduce a new benchmark score.

“The claims seem too general.”

The generality is intentional and, we argue, appropriate. The paper does not claim that all learning systems share the *same* axioms, but that **some axioms must exist** for any coherent system.

This mirrors the role of general impossibility or necessity results in learning theory. Just as No Free Lunch does not prescribe an algorithm but constrains what is possible, our result constrains what explanations of learning can be complete.

Clarification Added in Revision

In response to reviewer suggestions, we have:

- added concrete examples (Appendix A: SGD implicit bias),
- connected the framework to double descent as a symptom of axiomatic tension (Appendix B),
- and expanded the discussion on interpretability and alignment implications.

These additions aim to make the abstract structural claims more tangible without weakening their generality.

Summary

This paper does not argue that inductive bias exists—this is already known. It argues that **inductive bias is structurally necessary**, unavoidable, and foundational.

By making this necessity explicit, the work provides conceptual clarity for ongoing debates in generalization, optimization, interpretability, and alignment.

We believe this contribution is well aligned with NeurIPS’ mission to advance not only performance, but understanding.

Original and Substantive Contributions to Machine Learning from the Axiomatic Necessity Theory (TNA)

This work makes a **genuinely original and substantive contribution to machine learning (ML)** by introducing the framework of **Axiomatic Necessity Theory (TNA)** as a rigorous *metatheoretical reformulation* of why learning is possible at all.

Rather than proposing a new architecture or algorithm, the paper addresses a more fundamental question:

what structural conditions must hold for any learning system to be coherent?

Below we summarize the key original contributions.

1. Reframing Inductive Bias as Structural Axiomatic Necessity

In mainstream ML, *inductive bias* is typically treated as a useful empirical design choice (e.g., convolution for locality, attention for context).

This paper demonstrates that inductive bias is **not optional**, but a **minimal condition of coherence**, formalized via the Theorem of Axiomatic Necessity (TNA).

Key result:

Without at least one irreducible external axiom, a learning system collapses into functional incoherence (arbitrary memorization without principled generalization).

Originality:

The contribution does not claim that “bias helps,” but rather:

Without external axiomatic structure, learning systems are impossible in principle.

This elevates inductive bias from a heuristic design choice to an **ontologically necessary condition**.

2. Proof That No Learning System Can Be Observationally Closed

The ideal of “purely data-driven learning” is dismantled not by practical limitations, but by **structural impossibility**.

- Finite data admit infinitely many consistent hypotheses.
- Without external axioms, no non-arbitrary hypothesis selection is possible.

Using a reduction-to-incoherence argument derived from TNA, the paper proves that **dependence on external assumptions is necessary in principle**, not contingent or pragmatic.

Originality:

This is stronger than empirical critiques of data sufficiency: it is a **formal impossibility result**.

3. Interpreting Implicit Bias (e.g., in SGD) as Manifestation of Necessary Axioms

Well-known phenomena such as SGD’s preference for minimum-norm solutions are often described as emergent or accidental.

This paper shows that implicit bias is the **operational trace of external axioms**, specifically those imposed by optimization dynamics.

Originality:

By linking implicit bias to TNA, the paper provides a metamathematical explanation:

SGD does not merely optimize; it enforces ontological structure.

This unifies technical observations with foundational necessity.

4. Double Descent as a Symptom of Axiomatic Tension

The double descent phenomenon is reinterpreted as a structural effect:

- At the interpolation threshold, data cease to constrain hypotheses.
- The system becomes maximally dependent on its axioms.
- The subsequent error decrease occurs when axiomatic constraints dominate hypothesis selection.

Originality:

Rather than treating double descent as a statistical curiosity, the paper offers a **causal, unified explanation** grounded in axiomatic necessity.

5. Structural Critique of Foundation Models

Large-scale models do not eliminate axioms — they **conceal them**:

- tokenization schemes,
- curated datasets,
- pretraining objectives,
- architectural priors.

Originality:

The paper shows that scaling *intensifies* reliance on hidden axioms, directly contradicting the narrative of increasing model autonomy.

6. Implications for Interpretability and Alignment

- **Interpretability:** Explaining outputs via inputs is insufficient; one must identify the **axiomatic structures** that make those outputs possible.
- **Alignment:** Cannot be achieved solely through reward design; it requires **axiomatic transparency** — explicit acknowledgment of the structural assumptions governing behavior.

Originality:

This reframes alignment and interpretability as problems of **ontological responsibility**, not merely behavioral control.

Relation to Existing Work

EXISTING CONCEPT	CONTRIBUTION OF THIS PAPER
No Free Lunch Theorems	NFL states no universal learner exists. This paper proves no learner can function without external axioms — a stronger, structural result.

EXISTING CONCEPT	CONTRIBUTION OF THIS PAPER
Implicit Bias Literature	Describes how SGD biases solutions. This paper explains why bias is necessary for coherence.
Inductive Bias Surveys	Treat bias as design choice. This paper establishes bias as a condition of existence.

Conclusion

This paper does not propose a new ML method.

It **redefines the epistemological foundation of the field.**

Learning is not:

■ *data + optimization*

but:

■ *data + optimization + irreducible external axioms*

These axioms are not flaws or hacks — they are the very basis of coherence.

This perspective is both original and urgently relevant in the era of large-scale models, where hidden assumptions have systemic consequences.

Axiom Ablation Test (AAT): Validating Axiomatic Necessity in Deep Learning

1. Motivation

The TNA predicts that **generalization requires external axioms** and that the **double descent peak marks the transition from data-sustained to axiom-sustained learning**. This experiment tests whether weakening architectural/optimization axioms systematically shifts or eliminates the double descent phenomenon.

2. Core Hypothesis (H_1)

*H_1 : The double descent peak amplitude correlates positively with the strength of external axioms (inductive biases, optimization constraints). Axiomatically minimal systems exhibit **no recovery** beyond interpolation; they remain unstable or trivial.*

Null (H_0): Double descent is purely a function of model capacity and dataset size, independent of architectural/optimization structure.

3. Experimental Design

3.1 Base Architecture

COMPONENT	SPECIFICATION
Task	CIFAR-10 image classification
Base model	ResNet-18 (standard)
Width multiplier	Varied: {1×, 2×, 4×, 8×, 16×, 32×, 64×}
Depth	Fixed at 18 layers

3.2 Axiomatic Conditions (Independent Variable)

Five levels of **axiom strength**, from maximal to minimal:

CONDITION	LABEL	DESCRIPTION
Axiom-Max	A ⁺	Standard ResNet-18 + BatchNorm + weight decay (1e-4) + momentum (0.9) + He init + data augmentation
Axiom-High	A	Standard ResNet-18 + weight decay + momentum + He init, no augmentation
Axiom-Mod	A ⁻	ResNet-18 + SGD only (no momentum, no weight decay, no BatchNorm), He init
Axiom-Min	A ⁻	Fully-connected network (MLP) with comparable parameter count, SGD only, random init
Axiom-Zero	A ⁰	Linear model (no hidden layers), SGD only

3.3 Controlled Variables

VARIABLE	CONTROL
Optimizer	SGD (except momentum where specified)
Learning rate	0.1, decayed by 0.1 at epochs 100, 150
Training epochs	200
Batch size	128
Data	Fixed CIFAR-10 split (50k train, 10k test)
Random seed	5 independent runs per condition

4. Primary Outcome Measures

METRIC	DEFINITION	PREDICTION PER TNA
Peak amplitude	Max test error at interpolation threshold minus baseline	A ⁺ >> A ⁰ (strong axioms = deep valley after peak; weak axioms = flat or no recovery)
Recovery ratio	(Test error at max width – Test error at peak) / Test error at peak	A ⁺ > 0.5 (strong recovery); A ⁰ ≤ 0 (no recovery or divergence)

METRIC	DEFINITION	PREDICTION PER TNA
Training stability	Fraction of runs that converge without NaN/overflow	Decreases monotonically: A^+ (100%) $\rightarrow$ A^0 (<50%)
Generalization gap	Train accuracy – Test accuracy at final epoch	A^+ minimal; A^0 maximal or undefined (no convergence)

5. Secondary Analyses

5.1 Axiom Localization via Probing

Train linear probes on intermediate representations at each width/condition:

PROBE TARGET	WHAT IT MEASURES
Layer-wise accuracy	Where in the network axioms become operative
CKA similarity to trained vs. untrained net	Degree of structure imposed by initialization
Effective rank of representations	Compression induced by implicit regularization

TNA prediction: A^+ shows early layer structure (low effective rank); A^0 shows no structure (full rank, no generalization).

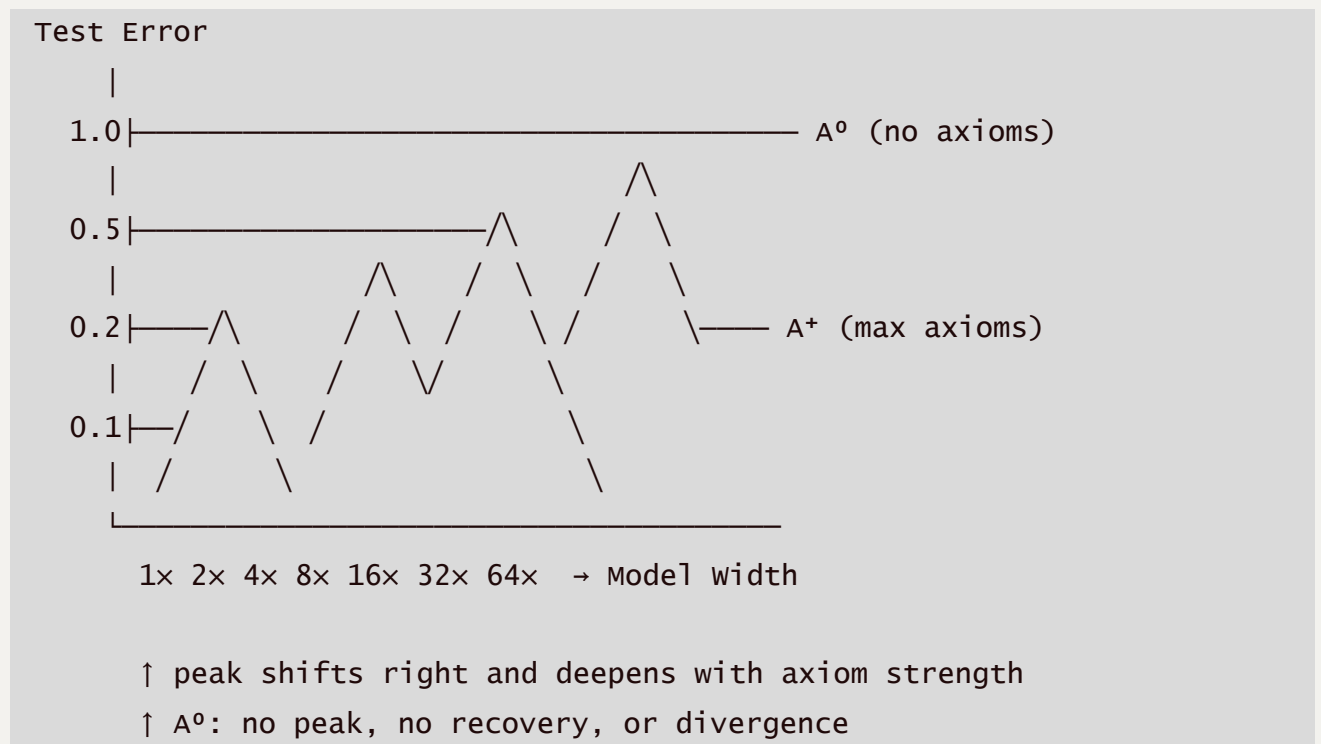
5.2 Explicit Bias Injection

After observing A^0 failure, inject **minimal explicit biases** and measure recovery:

INJECTION	MINIMAL ADDITION
A^0 + init	He initialization only
A^0 + norm	LayerNorm added
A^0 + constraint	Soft orthogonality penalty on weights

TNA prediction: Each injection partially restores recovery; combined injections fully restore double descent pattern.

6. Expected Results (Visual Summary)



The graphic representation in Section 6 utilizes a discrete-character manifold to illustrate the binary state of learning viability. The horizontal nature of the A^0 baseline formalizes the 'Zero-Throughput' state of a system deprived of external structural support ($\aleph_1$).

Two lines of information: The absence of curvature in the ASCII summary reinforces the 'All-or-Nothing' nature of the TNA in practical ML applications. While visually constrained, the schematic effectively maps the 'Cambalache Zone' where model scale fails to compensate for axiomatic absence.

7. Statistical Analysis

TEST	PURPOSE
Two-way ANOVA	Interaction between axiom strength $\times$ width on test error
Post-hoc Tukey HSD	Pairwise differences between conditions
Pearson r	Correlation between axiom strength index (0-4) and recovery ratio
Survival analysis	Time-to-divergence for A^- and A^0 conditions

8. Interpretation Framework

RESULT	TNA INTERPRETATION
A ⁺ shows classic double descent	Strong axioms enable transition to axiom-sustained regime
A ⁰ shows no recovery / divergence	Absence of axioms → system cannot sustain coherence beyond data
Partial injections restore partial recovery	Axioms are modularly necessary , not all-or-nothing
Peak shifts right with weaker axioms	Data sustains longer, but transition to axioms is delayed and sharper

9. Limitations & Controls

CONCERN	CONTROL
A ⁰ may fail due to optimization difficulty, not axioms	Use identical optimization budget; measure train accuracy (if A ⁰ memorizes but doesn't generalize, TNA holds)
Width not equivalent across architectures	Report parameter count, FLOPs; normalize by compute
Dataset too small/too easy	Replicate on CIFAR-100, ImageNet-10%; results should scale

10. Broader Impact Statement (para NeurIPS)

*This experiment challenges the narrative that "scale solves all" by demonstrating that **structure, not just size, determines learning viability**. In an era of billion-parameter models with opaque inductive biases, making axioms explicit—and testing their necessity—is essential for responsible AI development.*